NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

METRIC
MIL-PRF-24698(SH)
12 February 1988

PERFORMANCE SPECIFICATION

INFRARED THERMAL IMAGING SYSTEMS

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements for portable infrared thermal imaging systems (IRTIS) for use on board Naval ships for inspection and survey of electrical equipment.

1.2 Type. The IRTIS type designations shall be as follows:

A - self contained, directed viewing (through scanner) (see 3.2.1)
B - modular, remote viewing (separate scanner) (see 3.2.2)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.

MIL-T-28800 - Test Equipment for Use with Electrical and Electronic Equipment, General Specification for.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

FEDERAL

FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF COMMERCE

National Bureau of Standards Report NBSIR 85-3131.

(Copies of specifications, standards and other Government documents in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Device description. The IRTIS shall be a shipboard portable, opto-electronic, or electrical imaging system that will provide a real time image of infrared radiation.

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3.2.1 Type A. The type A IRTIS shall be a self-contained, directed viewing system. The power supply shall be provided either separate from or integral to the system. Controls shall be mounted directly to the system. The type A IRTIS will be used as an inspection or repair tool by shipboard personnel to inspect possible problem areas and verify repair of problems found during survey of shipboard electrical equipment.

3.2.2 Type B. The type B IRTIS shall be completely man-portable and shall display temperature, either directly or indirectly on the monitor. Controls shall be mounted directly to the monitor. The subassemblies are identified below. The subassemblies may be combined; however, the scanner shall be a separate subassembly. The type B IRTIS will be used to perform electrical surveys aboard ship. The type B IRTIS shall measure temperature and provide an instant developing record of the thermal image of the electrical equipment.

3.3 Subassemblies description.

3.3.1 Scanner unit. The scanner unit shall contain one or more detectors that sense infrared radiation. The scanner shall provide a persistent image to the monitor without the need of scanner motion.

3.3.2 Monitor unit.

3.3.2.1 Type A. The type A IRTIS shall produce a thermal image of the object being viewed by the scanner on a monitor. The monitor unit shall be updated at least 25 times a second for a real time image.

3.3.2.2 Type B. The type B IRTIS shall produce a thermal image of the object viewed by the scanner on a monitor. The monitor shall be updated at least 25 times a second for a real time image. The monitor unit shall provide sufficient information, either in analog or digital form, such that temperature differences and absolute temperatures can be determined. The real time frame image shall have no greater than 4 to 1 interlacing. The real time frame image shall contain at least 240 lines with display size at least 2 by 2 inches. The monitor shall be supplied with a camera mount fixture.

3.3.3 Power supply (battery pack). The power supply shall provide stable power for the operation of the IRTIS for a minimum of the continuous hours between charging intervals. It shall be recharged using the battery charger specified in 3.3.4.

3.3.4 Battery charger. The battery charger shall be supplied which is able to recharge the battery pack within a 18-hour period using a type 1 power source (nominal 115 volts alternating current (Vac), 60 hertz (Hz)) (see 4.7.2). The battery charger shall have an insulation resistance greater than 10 megohms (see 4.7.10).

3.3.5 Carrying case. Carrying cases shall protect the IRTIS during storage and in transit. The type A IRTIS shall be limited to one carrying case; the type B IRTIS shall have either one or two carrying cases. The carrying case shall be a hinged container with a cushion interior. Cutouts shall be provided for each separate component including: the basic unit, the battery pack (if separate), the battery charger, and a technical manual.

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3.3.6 Carrying harness. The type B IRTIS shall have a carrying harness that shall support and evenly distribute the entire weight of the scanner, monitor, battery pack, and camera accessory, on the operator. The harness shall not restrict the movement of the operator.

3.3.7 Camera mount. The type B IRTIS shall be supplied with a camera mount for photographic capability. The mount shall securely attach a camera that produces an instant, hard copy photograph of the displayed image on the IRTIS monitor. The camera mount shall be constructed so as to allow the operator to view the real-time displayed image in either direct sunlight or in a direct overhead lighting of 200 foot candles incident upon the monitor.

3.3.8 Camera. The type B IRTIS shall generate an instant photograph of the infrared thermal image displayed on the monitor. The IRTIS shall be constructed so that either the operator can view the image on the monitor while the photograph is being exposed, or so that the desired image shall be maintained, so that the operator is certain of the image being recorded.

3.3.9 Control unit. The control unit shall include, as a minimum, the following controls: on or off, brightness, contrast, and focus. In addition, type B IRTIS shall also contain controls for temperature measurement and polarity.

3.3.10 Interconnecting cable. The type B IRTIS shall have an interconnecting cable which separates the monitor from the scanner a distance of at least 100 feet.

3.4 Physical requirements.

3.4.1 Size and weight. The size and weight of the IRTIS shall be kept to a minimum consistent with good engineering design. The shape and weight of the load shall not interfere with the operator's:

(a) Length of step.
(b) Head movement.
(c) Ability to see where feet are placed when walking.
(d) Maintenance of normal posture.

3.4.1.1 Size. The IRTIS shall be one-man portable and easily harnessed to the body. The harnessed equipment shall permit freedom of movement through a raised circular opening with a diameter of 25 inches and a depth of 20 inches. Also, the type B scanner length, including lens, shall not exceed 12 inches.

3.4.1.2 Weight.

3.4.1.2.1 Type A. The total weight of the type A IRTIS shall not exceed 20 pounds.

3.4.1.2.2 Type B. Total weight of the type B IRTIS shall not exceed 35 pounds. The scanner weight shall not exceed 7.5 pounds. The total weight includes the weight of the scanner, monitor, battery pack, camera, harness, and shoulder strap assembly.

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3.5 Performance requirements.

3.5.1 Temperature range. The IRTIS shall produce and discern an infrared image of objects within a minus 10 to 300 degrees Celsius (°C) range.

3.5.2 Temperature measurement accuracy.

3.5.2.1 Type A. The type A IRTIS shall enable an operator with 20/20 vision (or 20/20 corrected vision) to discern, through shades of illuminance, a temperature difference between two blackbody sources where one source is 2 ± 1°C greater than the other (see 4.7.1.1).

3.5.2.2 Type B. The type B IRTIS shall measure object temperatures to within plus or minus 1 percent of the absolute temperature (in kelvins, K) of the reading (see 4.7.1.2). Temperature measurement shall be such that any object within a particular frame can be measured without movement of the scanner.

3.5.3 Temperature repeatability and battery pack. The type B IRTIS shall be repeatable to within plus or minus 0.5 percent of the absolute temperature (see 4.7.3).

3.5.4 Minimum resolvable temperature difference (MRTD).

3.5.4.1 Type A. The MRTD for the type A system shall be less than or equal to 0.2°C (see 4.7.4).

3.5.4.2 Type B. The MRTD for the type B system shall be less than or equal to 0.15°C (see 4.7.4).

3.5.5 Uniformity. The uniformity test as specified in 4.7.5 shall be conducted both horizontally and vertically. The average deviation determined by this test shall be no more than 6 percent.

3.5.6 Field of view.

3.5.6.1 Type A. The field of view of the type A IRTIS shall be at least 6.0 degrees horizontally by 6.0 degrees vertically (see 4.7.6).

3.5.6.2 Type B. The field of view of the type B IRTIS shall be at least 20 degrees horizontally by 20 degrees vertically (see 4.7.6).

3.5.7 Focal range. The IRTIS shall have a minimum focal range of 18 inches or less to infinity (see 4.7.7).

3.5.8 Inclination. The IRTIS shall operate at any inclination or orientation with no performance degradation (see 4.7.8).

3.5.9 Enclosure. The IRTIS shall have no evidence of water in the case following the testing specified in 4.7.9.

3.5.10 Storage temperature. The complete IRTIS, shall meet the accuracy requirements of 3.5.2, following storage at any temperature specified in 4.7.11.

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3.5.11 Operating temperature. The complete IRTIS, shall meet the accuracy requirement of 3.5.2 when operated over an ambient temperature range of minus 7 to 50°C (see 4.7.12).

3.5.12 Humidity. The IRTIS shall operate at humidity levels of at least 95 percent without condensation (at temperatures up to 50°C). There shall be no evidence of performance degradation or corrosion of any parts after testing (see 4.7.13).

3.5.13 Vibration. The IRTIS shall be tested in the carrying case and shall show no evidence of improper operation, failure, or damage (see 4.7.14).

3.5.14 Transit drop. The IRTIS shall be tested in the carrying case and shall show no evidence of improper operation, failure, or damage (see 4.7.15).

3.5.15 Bench handling. Subassemblies of the IRTIS shall show no evidence of improper operation, failure, or damage as a result of the bench handling (see 4.7.16).

3.5.16 Leakage current. The leakage current to the case shall not exceed 3.6 milliamps (mA) (see 4.7.17).

3.5.17 Electromagnetic interference (EMI). The IRTIS shall conform to the following requirements of MIL-STD-461: RE02 and RS03. The IRTIS shall meet the requirement of 3.5.2 (see 4.7.18).

3.6 Markings. Designations and markings shall conform to MIL-T-28800.

3.6.1 Warning markings. A warning marking shall denote the location of each safety (personnel) hazard, its nature, and its extent. Warning markings shall be in accordance with MIL-T-28800. In addition, IRTIS equipment with detectors made with mercury (for example, mercury cadium teluride) shall be marked in accordance with MIL-E-17555 and the hazard material marking documents specified therein.

3.6.1.1 Material safety data sheet (MSDS). The contracting activity shall be provided a MSDS at the time of contract award. The MSDS is form OSHA-20, found as part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

3.7 Maintainability. Operator level replaceable parts shall be accessible for corrective maintenance. Replacement of these parts shall not require removal from the equipment case and shall not violate the calibration of the unit. Intermediate level corrective maintenance actions shall consist of calibration, troubleshooting, and repairing, to the major subassembly level for electrical components, and to the component level for mechanical components.

3.7.1 Calibration. When specified in the contract or order, a calibration test shall be prepared (see 6.2.2).

3.7.2 Accessibility. The IRTIS shall be constructed so that the chassis may be removed from its enclosure with the use of ordinary tools. Detachable cords, spare fuses, and other accessories required for IRTIS operation and maintenance shall be stored in the carrying case. The IRTIS shall be constructed

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so that subassemblies and chassis compartments can be removed without removing any other hard wired subassembly or component. The IRTIS shall permit adjustments to be made without removing any component, printed circuit cards, or subassembly.

3.8 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The contractor shall be responsible for notifying the Government contract office of any failure during testing.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 of this specification. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality inspection program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in table I. The examinations and tests shall be performed in the order listed, except that the electromagnetic interference (EMI) test may be conducted on a separate sample and may be performed prior to, concurrent with, or after the other tests. When specified in the contract or order, a first article test report and procedure shall be prepared (see 6.2.2).

4.3.1 Samples for first article inspection. One IRTIS of each lot (see 4.3.2) shall be subjected to first article inspection.

4.3.2 Inspection lot. An inspection lot shall consist of all IRTISs of the same type, produced under essentially the same conditions, and offered for delivery at the same time.

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4.3.3 Failures. If the sample fails to pass first article inspections, the Government contract office shall be notified. The contractor shall identify cause of failure and take corrective action on the materials or process, or both, as warranted, and on all units of the products which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Shipment of the product shall not be initiated until corrective action has been taken. After the corrective action has been taken, first article inspection shall be repeated on another sample unit (all tests and examinations or the test which the original sample failed, at the option of the Government contract office). Final acceptance and shipment shall be withheld until the first article inspections have shown that corrective action was successful. All failure information shall be reported to the Government contract office.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table I in the order listed.

TABLE I. Examination and tests.

Test	Requirement	Test	First article	Quality conformance
General examination	3.2, 3.3, 3.4, 3.5.1	4.5	X	X
Temperature measurement accuracy	3.5.2	4.7.1	X	X
Battery charging time	3.3.4	4.7.2	X	
Temperature repeatibility and battery pack test	3.5.3, 3.3.3	4.7.3	X	
Minimum resolvable temperature difference	3.5.4	4.7.4	X	
Uniformity	3.5.5	4.7.5	X	
Field of view	3.5.6	4.7.6	X	
Focal range	3.5.7	4.7.7	X	
Inclination	3.5.8	4.7.8	X	
Enclosure	3.5.9	4.7.9	X	

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TABLE I. Examination and tests. - Continued

Test	Requirement	Test	First article	Quality conformance
Insulation resistance	3.3.4	4.7.10	X	X
Storage temperature	3.5.10	4.7.11	X	
Operating temperature	3.5.11	4.7.12	X	
Humidity	3.5.12	4.7.13	X	
Vibration	3.5.13	4.7.14	X	
Transit drop	3.5.14	4.7.15	X	
Bench handling	3.5.15	4.7.16	X	
Leakage current	3.5.16	4.7.17	X	
Electromagnetic interface	3.5.17	4.7.18	X	

4.4.1 Sampling for quality conformance inspection. IRTISs offered for delivery to the Government shall be subjected to the quality conformance inspection listed in table I.

4.4.2 Failures. If an IRTIS fails the quality conformance inspection, the unit shall be rejected and the contractor shall replace the unit.

4.5 General examination. The IRTIS shall be subjected to a thorough examination to ascertain that the material, finish, workmanship, construction, assembly, dimensions, and marking conform to the requirements of this specification and are in accordance with the contractor's approved drawings. Examinations shall be limited to those that may be performed without disassembling the IRTIS in such a manner that its performance, durability, or appearance would be affected. Examination shall also include a check of all adjustments and controls while the IRTIS is operational.

4.6 Test conditions. Except where the following factors are the variables, the tests in this specification shall be performed with the equipment and instrumentation operating under the following conditions:

(a) Ambient temperature shall be 24 ± 6°C.
(b) Relative humidity shall be 50 ± 10 percent.

4.7 Performance tests.

4.7.1 Temperature measurement accuracy.

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4.7.1.1 Type A (operational check). The operator having 20/20 vision or 20/20 corrected vision shall view the infrared thermal image of two blackbody sources, with equal emissivities and surface areas, concurrently. One blackbody shall have a temperature of 2 ± 1°C greater than the other which is held at 30 ± 1°C. Performance shall be as specified in 3.5.2.

4.7.1.2 Type B. The type B IRTIS test set-up shall be similar to that specified in NBSIR 85-3131. Target pattern number six shall be mounted perpendicular to and centered in the line of sight of the IRTIS scanner. The scanner field of view (FOV) shall include at least 50 percent of the background plate, but shall not extend outside the background plate boundary. A blackbody temperature source shall be used to provide a variable reference temperature (referred to as object temperature in NBSIR 85-3131). The reference temperature shall be varied from 0 to 100 percent of the span of the IRTIS at 20 percent intervals. At each of the six reference temperatures, the contractor's operating procedures shall be used to obtain the measured object temperature. Performance shall be as specified in 3.5.2.

4.7.2 Battery charging time. The IRTIS shall be operated until the battery pack is drained of power. The battery pack shall then be connected to the battery charger. The battery charger shall be energized for no more than 18 hours. The battery pack shall then be connected to the IRTIS and the temperature repeatability and battery pack test of 4.7.3 shall be performed.

4.7.3 Temperature repeatability and battery pack test. Prior to the start of this test the battery pack shall be fully charged (indicator on full). The IRTIS, excluding the battery charger, shall be placed in a temperature chamber in the non-operating condition. The chamber temperature shall be raised to 50 ± 3°C within 1 hour and allowed to stabilize. The IRTIS shall then be operated for a period of 2 continuous hours during which the temperature measurement accuracy test (see 4.7.1) shall be performed three times. The temperature test shall be performed at the beginning of the period, after 1 hour, and at the end of the period, with the completion of the test timed to coincide with the end of the 2 hour period. The IRTIS performance shall meet the requirements of 3.3.3 and 3.5.3.

4.7.4 Minimum resolvable temperature difference (MRTD). The MRTD test setup for the type A and B IRTIS shall be that shown in NBSIR 85-3131. The scanner shall view one of the five target patterns specified in 85-3131. The scanner shall be positioned at a distance such that the scanner is viewing a spatial frequency of 0.10 cycle per milliradian and the horizontal field of view subtends at least 1 meter in object space. The IRTIS shall be subjected to the following MRTD procedure:

(a) The IRTIS shall be aligned, both optically and thermally, by adjusting the necessary controls for a blackbody source temperature of 20 ± 2°C below ambient.
(b) The blackbody source temperature shall be increased until the bar pattern changes from black to white. The temperature (T1) of the blackbody source shall be recorded just as the bar pattern can be distinguished to be white.

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(c) The blackbody source shall then be decreased until the bar pattern changes from white to black. The temperature (T2) of the blackbody source shall be recorded just as the bar pattern can be distinguished to be black.
(d) Repeat steps (b) and (c) three additional times and record all T1 and T2 temperatures.
(e) Add all temperatures (T1) recorded in step (b).
(f) Add all temperatures (T2) recorded in step (c).
(g) Subtract step (f) total from step (e) total and divide the result by 4 to obtain the MRTD.

The MRTD shall be as specified in 3.5.4.

4.7.5 Uniformity. The IRTIS shall be tested in accordance with NBSIR 85-3131 and shall meet the requirements of 3.5.5.

4.7.6 Field of view. The IRTIS shall view a horizontal and vertical scale to determine the field of view of the system. The horizontal scale length, the vertical scale length, and the scanner to scale distance shall be measured (scanner to scale distance shall be a minimum of 1 meter). Determination of horizontal field of view (HFOV) and vertical field of view (VFOV) shall be as follows:

$$HFOV = 2 \arctan (HD/2L)$$

$$VFOV = 2 \arctan (VD/2L)$$

Where: HD = horizontal scale length within the field of view.
VD = vertical scale length within the field of view.
L = scanner to scale distance (from lens attachment).

The HFOV and VFOV shall meet the requirement specified in 3.5.6.

4.7.7 Focal range. The focal range test shall be performed by a person having 20/20 vision or vision corrected to 20/20 using corrective lenses. IRTIS shall meet the requirements of 3.5.7.

4.7.7.1 Minimum focal range. The IRTIS shall view a crosshair pattern, black on white background, at a scanner to pattern distance of 18 ± 1/4 inches. The crosshairs shall consist of lines 1/16 inch thick by 2 inches long. The pattern shall be centered in the field of view. The focus adjustment shall be set at infinity and slowly adjusted until the pattern is in focus. Adjustment in the same direction shall be continued until object is out of focus. Performance shall be in accordance with 3.5.7.

4.7.7.2 Maximum focal range. The IRTIS shall view an object with a sharply contrasted object to background emissivity, which is at least 150 feet from the scanner. The focus shall be adjusted to infinity and the observer shall ensure that object is in focus.

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4.7.8 Inclination. The IRTIS scanner shall be inclined from the normal attitude for a period of at least 1 minute at 90 degrees in four directions: forward, back, left, and right. In each position, a temperature measurement accuracy test shall be performed. IRTIS shall meet the requirements of 3.5.2 and 3.5.8.

4.7.9 Enclosure. The IRTIS, in its carrying case, shall be placed or mounted in a position typical of that for which it was designed. The surface upon which the equipment is placed or mounted (supporting surface) shall extend at least 3 feet beyond the equipment on all sides so that splashing may be produced by directing the water stream on that surface. The form of water flow shall be a coarse spray with a rate of water flow at least 15 gallons per minute, at a head pressure of at least 10 feet. A head pressure of 10 feet is defined as sufficient water pressure so that if directed straight up, the stream of water shall rise to a height of 10 feet. The distance from nozzle to the enclosure under test shall be approximately 6 feet. The maximum time of test shall be 5 minutes with approximately equal poritions of time for spray on each surface, including joints of the enclosure. The equipment shall conform to the requirements of 3.5.9.

4.7.10 Insulation resistance. The insulation resistance of the unpowered battery charger shall be determined by applying 50 volts direct current (dc) between each of the power supply leads and the case or cabinet. The current corresponding to the 50 volts dc shall be measured and the insulation resistance calculated. The IRTIS shall conform to the requirements of 3.3.4.

4.7.11 Storage temperature. The complete IRTIS, including the battery charger shall be placed in a temperature test chamber and shall remain in the non-operational mode during this test. The test shall subject the IRTIS to five complete temperature cycles, each of which is 48 hours in duration. Each cycle shall consist of the IRTIS being subjected to the different temperatures listed in table II, performed in the sequence listed. After completion of the fifth temperature cycle, the IRTIS shall be removed from the temperature test chamber and be allowed to stabilize at 24 ± 6°C for at least 4 hours. A temperature measurement accuracy test shall then be performed. The IRTIS performance shall meet the requirements of 3.5.10.

TABLE II. Storage test cycle.

Step	Temperature condition	Step duration
1	Increase temperature from ambient to 63 ± 3°C	1 hour, maximum
2	Maintain temperature	22 hours, minimum
3	Decrease temperature from 63 ± 3°C to minus 24 ± 3°C	2 hours, minimum
4	Maintain temperature	22 hours, minimum
5	Increase temperature from minus 24 ± 3°C to 24 ± 3°C	1 hour, maximum

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4.7.12 Operating temperature.

4.7.12.1 High temperature. The complete IRTIS, excluding the battery charger, shall be placed in a temperature test chamber in the non-operating condition. The temperature shall be increased to 50 ± 3°C within 1 hour and allowed to stablize. The IRTIS shall be subjected to this temperature for at least 4 hours. After this minimum 4 hour period, the temperature measurement accuracy test (see 4.7.1) shall be performed while the IRTIS remains at 50 ± 3°C inside the temperature test chamber. The IRTIS shall then be de-energized. The test chamber shall then be lowered to 24 ± 3°C and allowed to stabilize for at least 4 hours. A second temperature measurement accuracy test shall then be performed. The IRTIS performance shall meet the requirement of 3.5.11.

4.7.12.2 Low temperature. The complete IRTIS shall be subjected to the same test procedure as 4.7.12.1 with the exception that the temperature test chamber shall be lowered to minus 7 ± 3°C instead of increased to 50 ± 3°C. The IRTIS shall meet the requirement of 3.5.11.

4.7.13 Humidity. Temperature transitions shall be made in 1 hour or less. Relative humidity need not be maintained during transition periods. The complete IRTIS, excluding the battery charger, shall be subjected to the following humidity cycle in a non-condensing condition and shall meet the temperature measurement accuracy requirement of 3.5.2.

(a) The IRTIS, in the non-operating condition, shall be placed in an environmental test chamber and the temperature shall be increased to 50 ± 3°C within 1 hour. The IRTIS shall be subjected to this temperature for a period of 2 hours.
(b) The chamber temperature shall be reduced to 25 ± 3°C, 50 ± 5 percent relative humidity after which the temperature measurement accuracy test (see 4.7.1) shall be performed.
(c) The chamber temperature shall be raised to 50 ± 3°C at 95 percent relative humidity minimum, and maintain for 16 hours. The chamber temperature shall be lowered to 25 ± 3°C at 50 percent relative humidity minimum, and maintain for 8 hours minus the transition time.
(d) Step (c) shall be repeated four additional times (five 24-hour cycles). During the second cycle a temperature measurement accuracy test shall be performed to coincide with the end of 16-hour (50 ± 3°C at 95 percent humidity) period.
(e) Upon completion of five 24-hour cycles, chamber temperature shall be reduced to 25 ± 3°C and 50 percent relative humidity maintained for 4 hours.
(f) After removal from the environmental chamber, temperature measurement accuracy test shall be performed and IRTIS thoroughly examined for any sign of deterioration.

4.7.14 Vibration. The IRTIS shall be tested in its carrying case in accordance with MIL-STD-167-1 for type I equipment. A temperature measurement accuracy test shall be performed at the completion of the vibration test; the requirements of 3.5.2 and 3.5.13 shall be met.

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4.7.15 Transit drop. The IRTIS shall be tested in accordance with procedure IV, method 516.3 of MIL-STD-810. A temperature measurement accuracy test shall be performed after the completion of this test; the requirements of 3.5.2 and 3.5.14 shall be met.

4.7.16 Bench handling. Each subassembly of the IRTIS shall be removed from the carrying case and subjected to the bench handling test as specified in MIL-STD-810, method 516.3, procedure VI. A temperature measurement accuracy test shall be performed at the completion of this test; the requirements of 3.5.2 and 3.5.15 shall be met.

4.7.17 Leakage current. The IRTIS shall be tested in accordance with MIL-T-28800 and shall meet the requirement of 3.5.16.

4.7.18 Electromagnetic interference EMI. EMI testing shall be in accordance with the test methods of RE02 and RS03 of MIL-STD-462 and shall meet the requirements of 3.5.17.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4).

5.1 Packaging. Infrared thermal imaging systems in their carrying case (see 3.3.5) shall be individually preserved level A, C or commercial, packed level A, B, C or commercial as specified (see 6.2.1) and marked in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. The IRTIS is intended to produce a thermal picture of shipboard equipment and detect thermal anomalies which may be indicative of potential or actual failures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) IRTIS type (see 1.2).
(c) IRTIS quantity required.
(d) Level of packaging required (see 5.1).

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.7.1	Calibration system description	DI-R-7064	----
3.8	Drawings, engineering and associated lists	DI-E-7031	----
4.3	First article inspection procedure	DI-T-4901	----
4.3	First article inspection report	DI-T-4902	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a

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product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets (MSDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313. In order to obtain the MSDS, federal acquisition regulation (FAR) clause 52.223-3 must be in the contract.

6.6 Subject term (key word) listing.

Blackbody
Directed viewing
Electromagnetic interference
Remote viewing

Preparing activity:
Navy - SH
(Project 5855-N280)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-24698(SH)	INFRARED THERMAL IMAGING SYSTEMS

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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